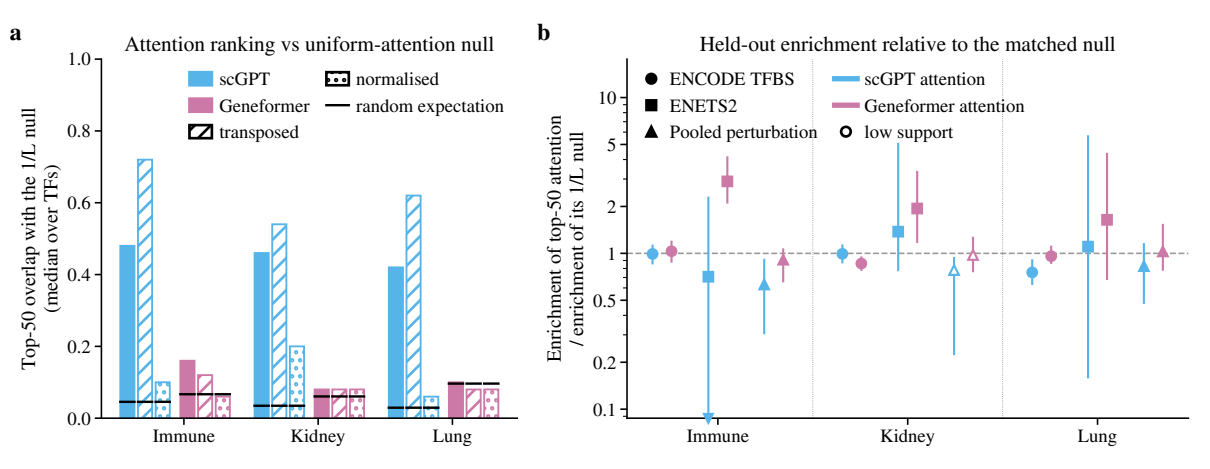

a
Attention ranking vs uniform-attention null
scGPT
Geneformer
transposed
normalised
random expectation
Top-50 overlap with the 1/L null
(median over TFs)
1.0
0.8
0.6
0.4
0.2
0.0
Immune
Kidney
Lung
b
Held-out enrichment relative to the matched null
ENCODE TFBS
ENETS2
Pooled perturbation
scGPT attention
Geneformer attention
low support
Enrichment of top-50 attention
/ enrichment of its 1/L null
10
5
2
1
0.5
0.2
0.1
Immune
Kidney
Lung